

Smart Multi-Zone Classroom Appliance Control

Using PIR-Based Presence Detection and Temperature-Driven Decision Logic

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Overview

This project presents a smart classroom power management system that automatically controls lights and fans based on occupancy and ambient temperature. Using PIR sensors for zone-by-zone presence detection and a DHT22 sensor for temperature-aware logic, the system ensures that only occupied areas of a classroom receive power — reducing electricity waste without any manual intervention.

The system is privacy-friendly (no cameras), low-cost, and built on the ESP32 microcontroller with IoT connectivity for long-term monitoring.

Problem Being Solved

- Lights and fans run continuously in empty or partially occupied classrooms, wasting electricity.
 - Single-sensor systems cannot tell which part of a room is occupied, powering everything unnecessarily.
 - Fixed timers turn off appliances too early when students are sitting quietly, causing discomfort.
 - Manual switching is unreliable — teachers and students often forget to turn off appliances when leaving.
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Objectives

- Automatically switch lights and fans ON or OFF based on real-time occupancy per zone.
 - Implement temperature-adaptive fan delay to balance comfort and energy savings.
 - Use inexpensive components (PIR, DHT22, ESP32) without cameras or complex hardware.
 - Enable IoT monitoring to track occupancy patterns and predict usage over time.
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How It Works

Zone-Based Occupancy Detection

Three PIR sensors are placed in the front, middle, and back of the classroom. Each sensor monitors its zone independently and reports to the ESP32 microcontroller. When motion is detected in a zone, only that zone's relay is activated — powering only the lights and fan for that area.

Temperature-Adaptive Fan Delay

After the last detected movement, the DHT22 sensor's temperature reading determines how long the fan stays on before switching off:

- Above 30°C → Fan stays ON for 8–10 minutes
- 27–30°C → Fan stays ON for 5–7 minutes
- Below 27°C → Fan stays ON for 2–3 minutes

Per-Zone Relay Control

Each classroom zone (Front, Middle, Back) has a dedicated relay channel. The ESP32 drives each relay through an NPN transistor with flyback protection diodes. When a zone is occupied or its delay hasn't expired, the relay stays on; otherwise it switches off automatically.

IoT Monitoring

The ESP32's built-in Wi-Fi uploads occupancy and temperature data to the cloud, enabling pattern analysis and predictive scheduling — so appliances can turn on slightly before classes begin.

System Block Diagram

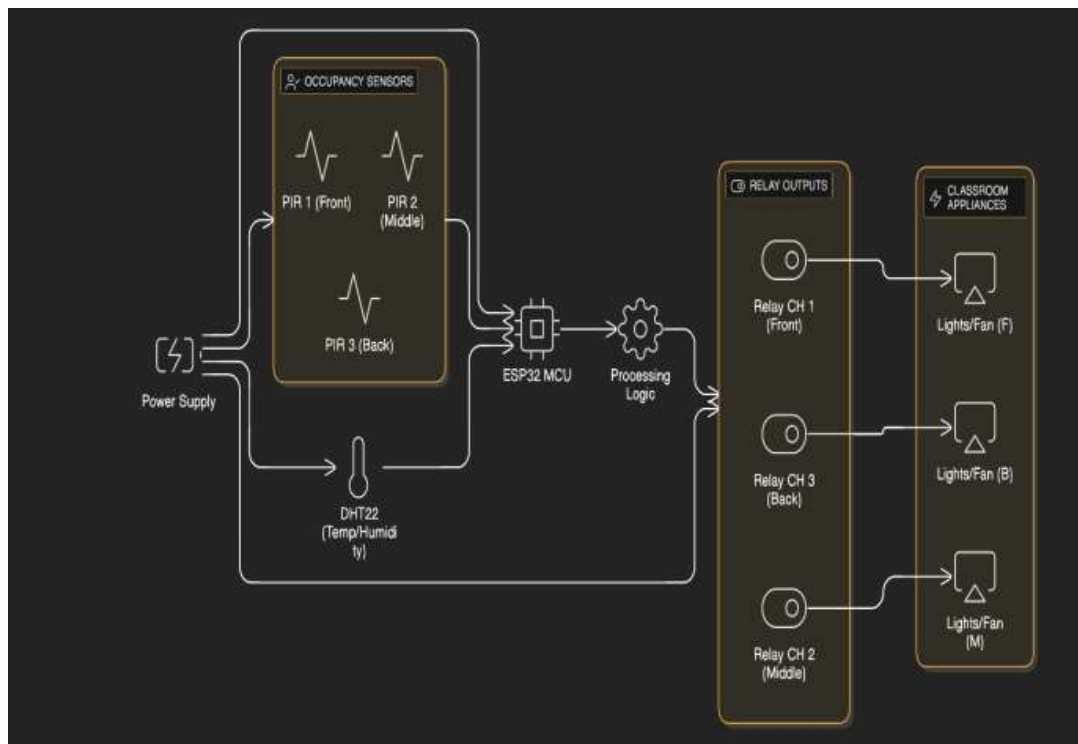


Figure 1: System architecture showing PIR sensors, DHT22, ESP32, relay channels, and classroom appliances

Power flows from a DC supply to the ESP32, sensors, and relay driver circuits. The three PIR sensors feed occupancy signals to the ESP32, and the DHT22 provides temperature and humidity readings. Based on these inputs, the ESP32 activates the appropriate relay channel (CH1 = Front, CH2 = Middle, CH3 = Back), each powering the lights and fan for that zone.

Circuit Diagram (Proteus Simulation)

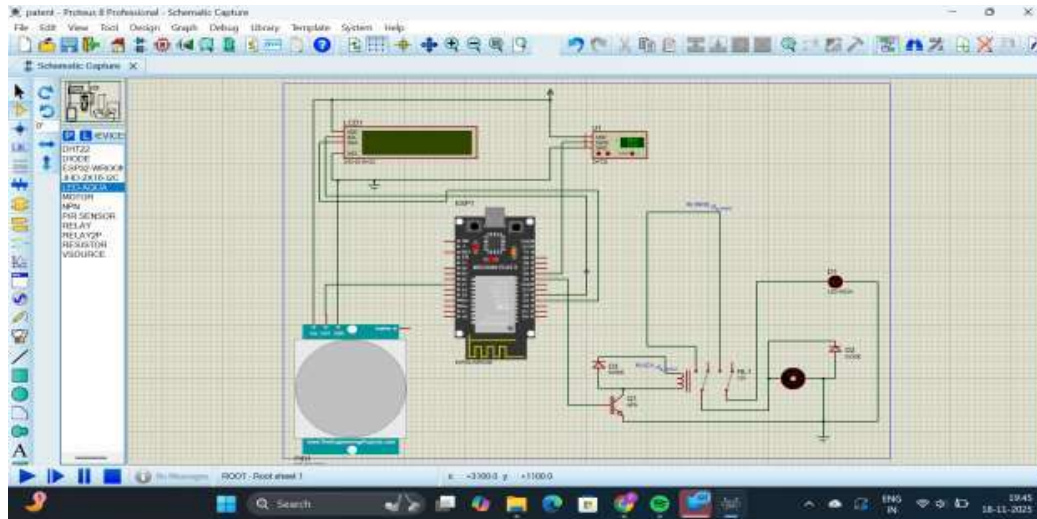


Figure 2: Full circuit schematic in Proteus — showing ESP32, PIR sensor, DHT22, relay driver, and load

Key Circuit Components

- ESP32 (ESP32-WROOM): Central controller. GPIO pins D5, D18, D19 used for sensors and relay control. Powered at 5V, internally regulated to 3.3V.
- PIR Sensor Module: 3-pin (Vcc, OUT, GND). Output goes HIGH on motion; signal fed to ESP32.
- DHT22 Sensor: Sends digital temperature and humidity readings via a single data pin.
- Relay Driver Stage: NPN transistor + base resistor + 1N4007 flyback diode. 12V DC relay coil switches AC/DC loads.
- Protection Diodes: D3 (flyback across coil) and D2 (flyback across motor load) protect against inductive voltage spikes.
- LCD Display (I2C): Displays temperature, humidity, zone status, and system messages via SCL/SDA lines.
- Power Supply: 12V for relays and loads; 5V (regulated) for ESP32, PIR, DHT22, and LCD.

Hardware Results

The system was assembled on a breadboard and tested in a real classroom environment. All functions performed as designed.

Test 1: No Motion Detected

When no movement is detected in any zone, all LEDs (tube lights) and fans remain off.

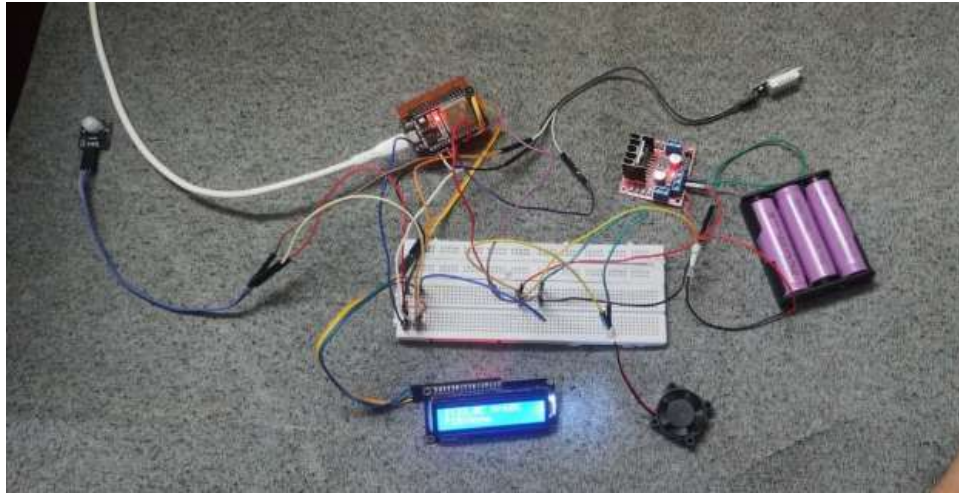


Figure 3: No motion — LED and fan both off

Test 2: Motion Detected, Temperature Below 27°C

When a PIR sensor detects motion, the LED for that zone turns on immediately. However, because the room temperature is below 27°C, the fan delay is short and the fan remains off.

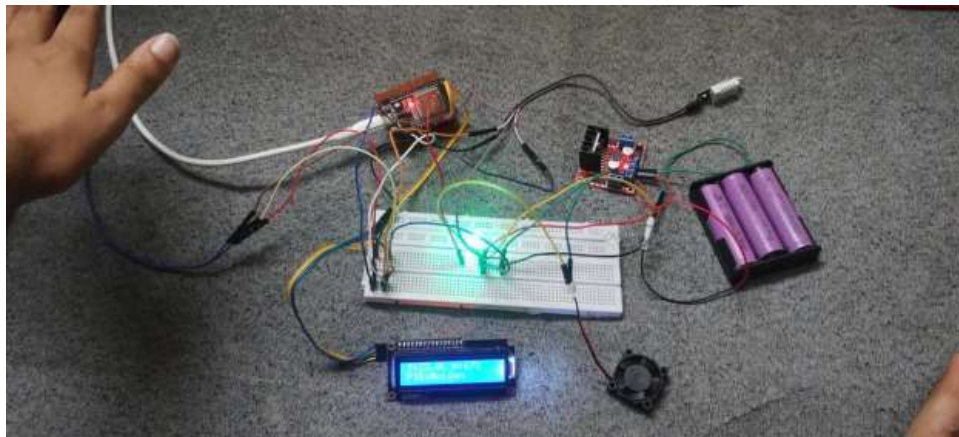


Figure 4: Motion detected, LED on — fan off because temperature < 27°C

Test 3: Temperature Above 27°C — Fan Activates

When the temperature rises above 27°C and motion is present, the fan turns on along with the LED. The LCD displays current temperature, humidity, and PIR status.



Figure 5: Temperature > 27°C — fan ON, LCD showing sensor readings

Advantages Over Existing Systems

Multi-Zone vs. Single-Sensor

Most existing systems use one PIR for the entire room, causing all appliances to run even when only one student is present. This system's three-zone architecture powers only occupied areas, significantly cutting waste.

Temperature-Adaptive Logic

Common motion-based systems use fixed timers, which often switch fans off while people are still present but sitting quietly. This system adjusts the delay dynamically based on ambient temperature, improving both comfort and efficiency.

Low-Cost and Scalable

The system uses inexpensive off-the-shelf components (PIR sensors, DHT22, ESP32, relays). Additional zones can be added without significant cost, making it suitable for classrooms of any size.

Privacy-Friendly

No cameras or biometric sensors are used. The system detects motion and heat only, with no ability to identify individuals.

Comparison with Prior Art

This system addresses specific limitations found in three relevant patents:

- US6415984B1 (PIR fan control): Works for a single fan with no zone support or humidity sensing. This system adds multi-zone mapping and DHT22 humidity-aware logic.
- US8900041B2 (temperature-based fan speed): No occupancy detection, so fans run even in empty rooms. This system adds PIR-based occupancy gating.

- IN2015DN02057A (occupancy sensor methods): Focuses on detection algorithms without low-cost microcontroller deployment or classroom zone control. This system delivers a complete, deployable ESP32-based solution with per-zone switching.
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Future Scope

- Cloud-based occupancy analytics to generate usage reports for energy auditing.
- Predictive scheduling: automatically pre-conditioning the room before a class starts based on historical patterns.
- Integration with institutional timetables for automated scheduling.
- Expansion to multi-room buildings using a networked cluster of ESP32 nodes.